

Department of Computer Engineering
National Institute of Technology Kurukshetra
 B.Tech (IT and MC), ITPC-206/DBMS, Mid Term (II) Exam, 4th Sem Sem, AY. 2025-26

Time Allowed: 50 Mins
 Max. Marks: 15.00

Date of Exam: 07 April, 2026
 Roll No: _____

Note: All the questions and their parts are compulsory; there is no choice. Answer each part of a question in the same place.

Q1. Answer the following briefly:

- "Faculty guides Project, Student works on Project", but actually a ternary: FACULTY guides STUDENT on PROJECT. Map this into relational schema.
- Why SQL does not allow to compare an attribute value to NULL using '=' and '<>' operators? How this problem is resolved in SQL?
- For given set for FDs, $F = \{A \rightarrow BC, B \rightarrow D, CD \rightarrow E\}$, Is $A \rightarrow E$ implied? Justify. *Yes, transitive rule*

Q2. (a). NIT Kurukshetra needs to track student enrollment history in courses, including: Multiple attempts of same course, Grade improvement over time, When the grade was awarded (date), and Who taught the course (different faculty each semester). The schema design is as follow: enrolls (roll_no, course_code, grade, semester, year). However it is observed that, current schema loses information when a student repeats a course and cannot track which faculty taught which offering. You have assigned role to redesigned the current schema to support:

- Multiple attempts of same course by same student.
- Tracking grade history (original grade, improved grade).
- Linking to specific course offering (faculty, semester, year).
- Query: "Find students who improved their grade from F to A in any course".

Write down the details of your proposed new schema to support above operations.

(b). The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade. List the set of all tuples that must be additionally deleted to preserve referential integrity when the tuple (2,4) is deleted, also rewrite the final table records.

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

(c). Consider a relation R (A, B, C, D, E) and FD set $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A, AE \rightarrow BD\}$. This relation is decomposed into two sub relations $r_1(ABC)$ and $r_2(ADE)$. Check if the decomposition is dependency preserving or not. Also check whether it is Lossless or lossy decomposition. Discuss Briefly.

Q3. Consider the Company relational DB, where the primary key(s) are underlined.

employee (employee-name, gender, salary, city, country)
 company (company-name, city, country)
 works (employee-name, company-name)
 manages (employee-name, manager-name)

Give an equivalent SQL expression for each of the following queries:

- Find the names of all female managers who work for the company named 'Flipkart' but do not live in any city in the 'India'.
- Find Employees 'Who Earn More Than ALL Employees in 'Delhi'.
- For each company that has more than five thousand employees and located in at least 03 cities, retrieve the company name and number of its employees, who are making more than (INR) 15,000.
- For a given country, retrieve the names of companies with the lowest number of female employees.